ABHISHEK TRIPATHI

MSc Informatics | Cybersecurity | SOC | Python | Security Automation

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PROFESSIONAL SUMMARY

MSc Informatics student at the University of Delhi building a career in cybersecurity, with a focus on SOC operations, ethical hacking, security automation, Python, Linux, and networking. Hands-on interest in log analysis, threat detection, OSINT, MITRE ATT&CK, automation, and AI-assisted security workflows. Academic foundation in Electronics Science with additional experience across Python, AI/ML, RAG applications, web development, and client-facing digital work.

EDUCATION

University of Delhi, Institute of Informatics and Communication (IIC), South Campus — **2026 – Present**
MSc Informatics (First Year)

University of Delhi, Acharya Narendra Dev College — **2023 – 2026**
BSc (Hons.) Electronics Science

Maharishi Vidya Mandir Senior Secondary School — **2019 – 2023**
Senior Secondary Education

TECHNICAL SKILLS

Cybersecurity: SOC Operations, Security Monitoring, Log Analysis, Threat Hunting, OSINT, Ethical Hacking, MITRE ATT&CK, EDR (Endpoint Detection & Response)

Security & Automation: Python Automation, SIEM Concepts, Brute-Force Detection, Security Scripting

Systems & Networking: Linux, Windows, Networking Fundamentals, Nmap

Programming: Python, SQL, HTML, CSS, JavaScript, TypeScript

AI / Data: Machine Learning, Generative AI, RAG, Hybrid Search (BM25 + Embeddings), ChromaDB, Ollama, LangChain, Pandas

Development: FastAPI, React, Tailwind CSS, SQLite, Git/GitHub, REST APIs, n8n

Tools: Jupyter Notebook, VS Code, Kali Linux, Microsoft Excel

PROJECTS

AI-Powered SOC Analyst – Threat Detection & Triage Platform | github.com/muskabhishek/soc-ai-analyst

Tech: *Python, Pandas, Scapy, Wireshark/PCAP, MITRE ATT&CK, Splunk SPL, Sentinel KQL, FastAPI, Streamlit, SQLite, Ollama, GitHub Actions*

- Built a SOC L1 triage platform that ingests Linux SSH logs, Windows Security events and Wireshark packet captures, detecting attacks with 19 rules mapped to MITRE ATT&CK (brute force, password spraying, privilege escalation, encoded PowerShell, port scans, C2 beaconing, DNS tunnelling, remote shells).
- Validated on real attack data: an internet-facing server's SSH log (86,839 lines → 236 alerts, 0 on internal IPs) and 13 public PCAPs (4 of 7 attacks detected, 0 of 6 benign captures flagged); real traffic exposed a false positive and log-parsing gaps, both fixed.
- Built a case workflow (True/False Positive verdicts, audit trail, MTTA/MTTR, false-positive rate per rule), cross-source attacker correlation, and 19 triage playbooks with a 5W worksheet that pre-fills evidence and drafts the closing comment.
- Wrote the detections in Splunk SPL and Microsoft Sentinel KQL; every network alert carries a Wireshark display filter; a local LLM (Ollama) drafts incident reports with invented-IP checks; 25 automated tests in GitHub Actions.

EduAstra – Exam-Ready AI Study Assistant (RAG) | *Team Project, Acharya Narendra Dev College*

Tech: *Python, FastAPI, React, ChromaDB, BGE-M3, BM25, Ollama (Qwen2.5), PyMuPDF, Kaggle GPU*

- Built a hybrid retrieval (BGE-M3 dense + BM25) RAG system over NCERT textbooks, previous-year papers and CBSE marking schemes generating marks-scaled (2/5/10-mark) board answers; achieved 92.9% top-3 retrieval accuracy vs 85.7% dense-only and 82.1% BM25-only on a 28-question test set.
- Engineered automatic cloud-to-local LLM failover (Kaggle T4 GPU via Cloudflare tunnel → local Qwen2.5-3B) with health checks and mid-stream recovery, at zero API cost.
- Built a page-routing PDF extraction pipeline (prose, maths, scanned and handwritten pages with vision OCR), raising usable English page coverage from ∼17% to 37.8%.
- Developed a React + FastAPI web app with streamed answers (SSE), KaTeX maths rendering, source citations, chat history and dark mode.

CERTIFICATIONS

- Introduction to Generative AI, Google Cloud Skills Boost

ORGANIZATIONS & ACHIEVEMENTS

- Member, SPIE (International Society for Optics and Photonics)
- Bronze Medal, Olympiad